RL locomotion steering: hc-running-002

2026-10-22T16:33:53+00:00

Outcome

Recorded run status: no_validation_candidate.

All 24 primary-vector strength settings failed at least one usefulness gate across 120 validation conditions under the revised episode-level physical-failure criterion. The height vector at alpha=+0.05 raised torso height by 4.5048 cm (95% paired-episode CI 4.3673-4.6452 cm), with zero observed physical failures, contact or inversion, but reduced speed from 16.7955 to 14.6371 m/s (-12.85%), exceeding the predeclared 10% speed-preservation allowance. This is a causal posture effect on exploratory validation, not a useful confirmed intervention. Eight control settings passed validation gates; they are unconfirmed and do not establish specificity of contrastive extraction. No selection, confirmation or replication was performed. A fresh experiment will test the exact height-derived direction for useful speed control, with new validation and held-out episodes; the fitting label need not equal the useful outcome.

Validation selection: No selected intervention was recorded.

A gate pass is a pilot usefulness decision for one condition; it does not establish generalization across training seeds.

Question and method

Can adding a constant vector after the first actor ReLU change locomotion while preserving movement? The policy stays frozen. Vectors are episode-weighted differences of contrasting unsteered activations, normalized to centered activation RMS. The protocol includes random and shuffled-label controls on the same evaluation split when that stage is reached. No behavioral cloning is used.

Interpretation

Delta is treated minus baseline in the behavior's units. Speed/lateral use m/s, height uses m, turning uses rad/s, and effort is mean squared normalized action, not physical energy. Confidence intervals resample paired episodes. Validation selects strengths; confirmation and replication provide fresh evidence when recorded. Useful effects require the target threshold, a confidence interval excluding zero, and locomotion/quality checks. Control effects must be considered before attributing specificity to the extracted direction.

Fitting diagnostics

Fitting windows: episodes=64; total windows=576; eligible windows=564; eligible episodes=64; excluded nonfinite=0; excluded unhealthy=12; discarded tail steps=0; warmup=100; window=100; minimum speed fraction=0.5; minimum speed floor=7.327; healthy median speed=14.654; excluded slow windows=0; speed filter reason=project pilot heuristic: exclude finite healthy fitting windows at or below fraction times their median speed.

Centered activation RMS: 8.705. Convention: episode-balanced eligible timestep RMS about episode-balanced mean; full 100-step windows after 100-step warmup.

effort (mean squared action): status=fitted; high episodes=60; low episodes=57; high windows=144; low windows=144; contrast=0.031536; raw norm=0.43919; split-half cosine=0.89514.

height (m): status=fitted; high episodes=61; low episodes=61; high windows=141; low windows=141; contrast=0.044745; raw norm=1.2007; split-half cosine=0.97976.

speed (m/s): status=fitted; high episodes=61; low episodes=59; high windows=141; low windows=141; contrast=2.3066; raw norm=1.021; split-half cosine=0.98237.

Split-half cosine measures extraction agreement, not causal effectiveness. Full bin statistics and vector coordinates are retained in vector_diagnostics.json.

Research decisions

2026-09-07T23:05:48.788589+00:00: Removing near-stationary recovery windows prevents unwrapped rotation history from dominating the differences. A fitting-only half-median forward-speed floor should retain broad independent running contrasts. Evidence: reports/hc-classic-001-diagnostics.md; https://github.com/Farama-Foundation/Gymnasium/blob/v1.0.0/gymnasium/envs/mujoco/half_cheetah_v5.py.

2026-09-07T23:10:28.010858+00:00: Adopt inference-only buffer_size=1 loader override and audited cache migration before calibration. Evidence: inference-memory-audit.json: unchanged policy hash and byte-for-byte equality of every saved array for diagnostic seed 100000 and fitting seeds 101000 and 101010. Replay array allocation reduced from 308000000 to 1540 bytes..

2026-09-07T23:19:14.777484+00:00: Correct physical episode failure before selection or confirmation: an episode fails if its post-onset inversion or forbidden-contact fraction exceeds 5%, in addition to termination/prefix failure. Preserve pooled fractions and retain the existing maximum five-percentage-point increase in failure. Recompute all cached validation effects under this derived-analysis correction. Evidence: reports/hc-running-002-validation-inspection.md: one of ten episodes stalls on its head for400steps while pooled contact appears4.47%. The old74condition estimates remain in validation_pooled-quality-v1.json. No selection or confirmation exists..

Validation

120 recorded intervention conditions.

Vector / behavior	Alpha	Delta	95% CI	Pairs	Useful
speed / speed	-0.5	-10.88	[-11.968, -10.035]	10	no
speed / speed	-0.2	-5.747	[-7.0605, -4.9655]	10	no
speed / speed	-0.1	-4.3495	[-6.0546, -3.0911]	10	no
speed / speed	-0.05	-1.9767	[-3.5535, -1.1366]	10	no
speed / speed	0.05	-4.0003	[-5.1148, -3.0849]	10	no
speed / speed	0.1	-8.2289	[-10.484, -6.1884]	10	no
speed / speed	0.2	-11.409	[-13.248, -9.5488]	10	no
speed / speed	0.5	-14.55	[-15.004, -14.22]	10	no
speed_random0 / speed	-0.5	-13.912	[-14.678, -13.258]	10	no
speed_random0 / speed	-0.2	-10.658	[-12.717, -8.744]	10	no
speed_random0 / speed	-0.1	-6.6627	[-10.282, -3.2438]	10	no
speed_random0 / speed	-0.05	-0.10282	[-0.15059, -0.066691]	10	no
speed_random0 / speed	0.05	-1.1132	[-1.3913, -0.8554]	10	yes
speed_random0 / speed	0.1	-4.1331	[-5.4255, -3.4101]	10	no
speed_random0 / speed	0.2	-6.2346	[-6.3931, -6.0776]	10	yes
speed_random0 / speed	0.5	-13.493	[-14.766, -12.015]	10	no
speed_random1 / speed	-0.5	-14.715	[-15.371, -14.046]	10	no
speed_random1 / speed	-0.2	-8.8389	[-10.888, -6.7892]	10	no
speed_random1 / speed	-0.1	-8.2891	[-11.123, -5.4217]	10	no
speed_random1 / speed	-0.05	-0.13408	[-0.22555, -0.082921]	10	no
speed_random1 / speed	0.05	-0.64433	[-0.7366, -0.5647]	10	no
speed_random1 / speed	0.1	-1.7246	[-1.9342, -1.5711]	10	yes
speed_random1 / speed	0.2	-8.7111	[-10.388, -7.8019]	10	no
speed_random1 / speed	0.5	-14.453	[-15.037, -13.966]	10	no
speed_random2 / speed	-0.5	-14.457	[-15.316, -13.582]	10	no

Vector / behavior	Alpha	Delta	95% CI	Pairs	Useful
speed_random2 / speed	-0.2	-6.2464	[-8.0422, -4.8445]	10	no
speed_random2 / speed	-0.1	-2.5404	[-2.722, -2.3471]	10	yes
speed_random2 / speed	-0.05	-1.0334	[-1.1229, -0.94466]	10	yes
speed_random2 / speed	0.05	-1.1626	[-1.3582, -0.98082]	10	yes
speed_random2 / speed	0.1	-4.1617	[-6.643, -2.6235]	10	no
speed_random2 / speed	0.2	-7.7017	[-10.187, -5.3734]	10	no
speed_random2 / speed	0.5	-14.717	[-15.364, -14.025]	10	no
speed_shuffled / speed	-0.5	-16.062	[-16.2, -15.871]	10	no
speed_shuffled / speed	-0.2	-14.064	[-14.732, -13.451]	10	no
speed_shuffled / speed	-0.1	-7.7364	[-8.787, -7.0291]	10	no
speed_shuffled / speed	-0.05	-4.9478	[-6.6663, -3.6443]	10	no
speed_shuffled / speed	0.05	-3.6287	[-5.6744, -2.2027]	10	no
speed_shuffled / speed	0.1	-4.7923	[-4.9021, -4.692]	10	yes
speed_shuffled / speed	0.2	-9.9402	[-11.526, -8.5347]	10	no
speed_shuffled / speed	0.5	-15.833	[-15.96, -15.727]	10	no
effort / effort	-0.5	0.10612	[-0.044243, 0.25396]	10	no
effort / effort	-0.2	-0.21136	[-0.41411, -0.035961]	10	no
effort / effort	-0.1	0.0047763	[-0.011708, 0.015168]	10	no
effort / effort	-0.05	-0.067746	[-0.19745, 0.0093414]	10	no
effort / effort	0.05	-0.15014	[-0.24166, -0.10033]	10	no
effort / effort	0.1	-0.16233	[-0.22071, -0.13137]	10	no
effort / effort	0.2	-0.31979	[-0.44, -0.20473]	10	no
effort / effort	0.5	-0.17684	[-0.2908, -0.070533]	10	no
effort_random0 / effort	-0.5	-0.17966	[-0.33821, -0.038347]	10	no
effort_random0 / effort	-0.2	-0.022178	[-0.027144, -0.016421]	10	no
effort_random0 / effort	-0.1	0.012501	[0.006953, 0.016983]	10	no
effort_random0 / effort	-0.05	0.010626	[0.0098257, 0.011433]	10	no
effort_random0 / effort	0.05	-0.087288	[-0.16181, -0.032282]	10	no
effort_random0 / effort	0.1	-0.095529	[-0.17511, -0.053285]	10	no
effort_random0 / effort	0.2	-0.27357	[-0.39932, -0.16]	10	no
effort_random0 / effort	0.5	-0.27078	[-0.39756, -0.15039]	10	no
effort_random1 / effort	-0.5	-0.22416	[-0.35202, -0.089403]	10	no
effort_random1 / effort	-0.2	-0.078155	[-0.14809, -0.034796]	10	no
effort_random1 / effort	-0.1	-0.046841	[-0.0493, -0.044062]	10	no
effort_random1 / effort	-0.05	-0.069815	[-0.072032, -0.067148]	10	no
effort_random1 / effort	0.05	-0.074257	[-0.1626, -0.028276]	10	no
effort_random1 / effort	0.1	-0.062522	[-0.11819, -0.032946]	10	no
effort_random1 / effort	0.2	-0.16957	[-0.313, -0.044727]	10	no
effort_random1 / effort	0.5	0.080437	[0.027823, 0.14291]	10	no
effort_random2 / effort	-0.5	-0.21219	[-0.31504, -0.12167]	10	no

Vector / behavior	Alpha	Delta	95% CI	Pairs	Useful
effort_random2 / effort	-0.2	-0.25437	[-0.36722, -0.15294]	10	no
effort_random2 / effort	-0.1	-0.2124	[-0.35107, -0.087149]	10	no
effort_random2 / effort	-0.05	-0.25409	[-0.40374, -0.11173]	10	no
effort_random2 / effort	0.05	-0.050149	[-0.050944, -0.049288]	10	no
effort_random2 / effort	0.1	-0.083977	[-0.087673, -0.079713]	10	no
effort_random2 / effort	0.2	-0.11216	[-0.20242, -0.027723]	10	no
effort_random2 / effort	0.5	-0.30686	[-0.45486, -0.16301]	10	no
effort_shuffled / effort	-0.5	-0.4007	[-0.53581, -0.23729]	10	no
effort_shuffled / effort	-0.2	-0.3205	[-0.43991, -0.239]	10	no
effort_shuffled / effort	-0.1	-0.21208	[-0.26552, -0.18451]	10	no
effort_shuffled / effort	-0.05	-0.12516	[-0.12705, -0.12343]	10	no
effort_shuffled / effort	0.05	-0.095646	[-0.22004, -0.0050369]	10	no
effort_shuffled / effort	0.1	-0.076359	[-0.23708, 0.020999]	10	no
effort_shuffled / effort	0.2	-0.020479	[-0.11747, 0.063691]	10	no
effort_shuffled / effort	0.5	0.072554	[-0.10219, 0.19519]	10	no
height / height	-0.5	0.023742	[-0.08026, 0.10046]	10	no
height / height	-0.2	-0.13093	[-0.26522, 0.003618]	10	no
height / height	-0.1	-0.05372	[-0.16849, 0.04479]	10	no
height / height	-0.05	-0.044675	[-0.14405, 0.038453]	10	no
height / height	0.05	0.045048	[0.043673, 0.046452]	10	no
height / height	0.1	0.010394	[-0.074733, 0.06601]	10	no
height / height	0.2	0.080281	[0.078255, 0.082337]	10	no
height / height	0.5	-0.097293	[-0.21659, 0.017285]	10	no
height_random0 / height	-0.5	-0.053939	[-0.16523, 0.035621]	10	no
height_random0 / height	-0.2	0.034212	[-0.0078723, 0.065586]	10	no
height_random0 / height	-0.1	0.017082	[-0.033732, 0.047986]	10	no
height_random0 / height	-0.05	-0.00028536	[-0.0028792, 0.002258]	10	no
height_random0 / height	0.05	0.032571	[0.032116, 0.033033]	10	yes
height_random0 / height	0.1	0.049637	[0.048055, 0.051702]	10	no
height_random0 / height	0.2	0.066213	[0.062862, 0.068625]	10	no
height_random0 / height	0.5	0.084806	[0.07219, 0.097376]	10	no
height_random1 / height	-0.5	-0.17554	[-0.25841, -0.087529]	10	no
height_random1 / height	-0.2	0.006553	[-0.027155, 0.027704]	10	no
height_random1 / height	-0.1	0.027733	[0.024231, 0.031642]	10	no
height_random1 / height	-0.05	0.022621	[0.01889, 0.027462]	10	no
height_random1 / height	0.05	-0.0028807	[-0.0038593, -0.0012669]	10	no
height_random1 / height	0.1	0.0044741	[-0.036946, 0.036999]	10	no
height_random1 / height	0.2	-0.042715	[-0.12773, 0.038216]	10	no
height_random1 / height	0.5	-0.15758	[-0.26259, -0.044153]	10	no
height_random2 / height	-0.5	0.029053	[-0.018905, 0.067719]	10	no

Vector / behavior	Alpha	Delta	95% CI	Pairs	Useful
height_random2 / height	-0.2	-0.02546	[-0.11981, 0.049598]	10	no
height_random2 / height	-0.1	-0.012532	[-0.091081, 0.042867]	10	no
height_random2 / height	-0.05	0.01451	[0.0097428, 0.01842]	10	no
height_random2 / height	0.05	-5.2081e-05	[-0.00053692, 0.00057621]	10	no
height_random2 / height	0.1	0.023452	[0.012994, 0.031722]	10	no
height_random2 / height	0.2	0.057148	[0.049831, 0.063851]	10	no
height_random2 / height	0.5	0.016009	[-0.062058, 0.056782]	10	no
height_shuffled / height	-0.5	-0.2855	[-0.34432, -0.21156]	10	no
height_shuffled / height	-0.2	-0.1622	[-0.24145, -0.09309]	10	no
height_shuffled / height	-0.1	-0.0084491	[-0.083871, 0.032854]	10	no
height_shuffled / height	-0.05	0.035752	[0.033761, 0.038265]	10	no
height_shuffled / height	0.05	-0.002661	[-0.10628, 0.060543]	10	no
height_shuffled / height	0.1	0.00067204	[-0.095246, 0.075931]	10	no
height_shuffled / height	0.2	-0.082819	[-0.15242, -0.019885]	10	no
height_shuffled / height	0.5	-0.11124	[-0.22925, 0.0061281]	10	no

Unmeasured stages

No recorded measurements: confirmation, replication. No causal steering outcome is inferred for these stages.

Reproduction and provenance

Created: 2026-09-07T23:05:23.662755+00:00; code commit: not recorded.

Policy: farama-minari/HalfCheetah-v5-TQC-medium; algorithm: TQC; environment: HalfCheetah-v5.

Checkpoint revision: b4ce04da6f246f06ae4c4258b8ab39624e6600b4. SHA256: 8231919095d5a35a2b799e865e968feca535d5af3f40a4b092947c70a976b7e1.

Layer: actor.latent_pi.1; hidden dimension: 256; observation/action shapes: [17] / [6].

Seed partitions below are registered assignments, not completion counts.

Diagnostic: 16 seeds; range 100000-100015.

Fit: 64 seeds; range 101000-101063.

Validation: 10 seeds; range 110000-110009.

Confirmation: 30 seeds; range 120000-120029.

Replication: 30 seeds; range 130000-130029.

Configuration: torch_threads=1; rollout_workers=4; max_steps=1000; warmup=100; diagnostic_episodes=16; fit_episodes=64; validation_episodes=10; confirmation_episodes=30; replication_episodes=30; seed_offset=1e+05; fit_min_speed_fraction=0.5.

Software: numpy 1.26.4; torch 2.5.1+cpu; torchvision 0.20.1+cpu; stable-baselines3 2.4.1; sb3-contrib 2.4.0; gymnasium 1.0.0; mujoco 3.1.5; baukit 0.0.1.

Full exact seed lists, checkpoint metadata and the original specification remain in adjacent manifest.json. Full measured analysis is in results.json; extraction details are in vector_diagnostics.json. The report does not change these artifacts or rerun inference.

Sources

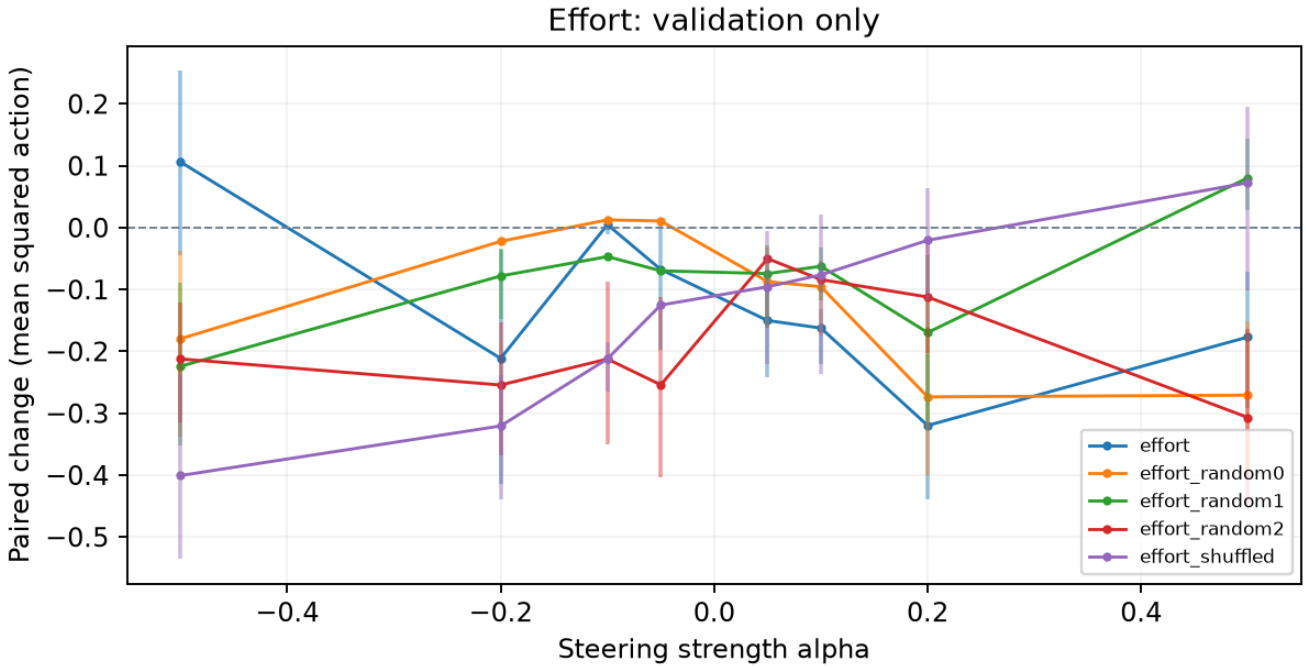
CAA method: https://arxiv.org/abs/2312.06681

Baukit hooks: https://github.com/davidbau/baukit/blob/9d51abd51ebf29769aecc38c4cbef459b731a36e/baukit/nethook.py

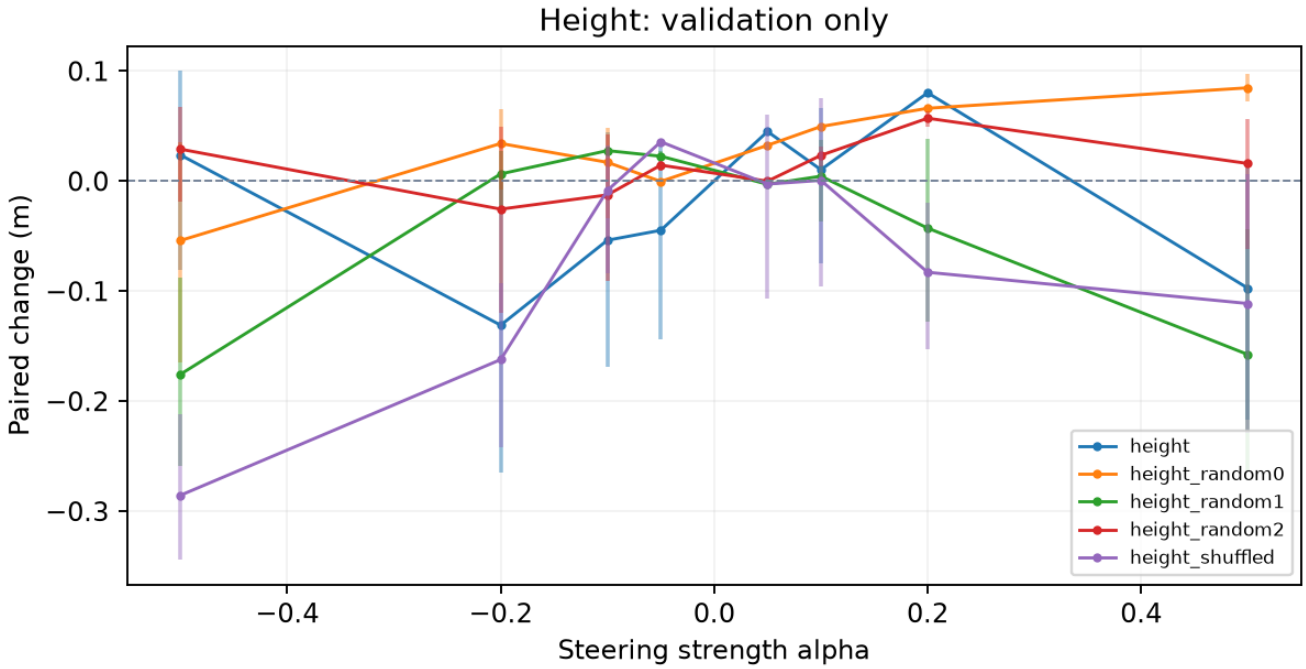
Environment and measurement references: docs/sources.md in the repository.

Reporting: https://docs.reportlab.com/reportlab/userguide/ch5_platypus/ and https://matplotlib.org/stable/users/index.html

Validation strength response: effort



Validation strength response: height



Validation strength response: speed

